THE ROLE OF EXPERIMENTATION IN CREATING AND SUSTAINING MOTIVATION IN DESIGN WORK

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ABSTRACT

The nature of design and development work requires self-directed proactive and persistent efforts from both the design team and other stakeholders. However, the question of how to enhance development motivation can be a daunting one to answer, as the literature is dispersed, partial answers being presented in multiple fields of research. This paper specifically examines the role that experimentation, a typical and widely supported approach to design, can play in awakening and sustaining development motivation. The paper reviews literature from several streams of research and discusses findings from a recent attempt to realize support for these mechanisms within a university setting, at the Aalto University Design Factory platform.

Keywords: new product development, motivation, experimentation

INTRODUCTION

New product development plays a large role in sustaining company performance and competitiveness (Dougerthy & Hardy 1996; Schilling & Hill 1998; Ernst 2002), and can act as a core mechanism in the knowledge creation of the whole organization (Nonaka & Takeuchi 1995). However, mere compliance does not suffice for the continuous and uncertain development work – proactive and persistent efforts are required from the development team and stakeholders. In addition to the complex problems requiring a variety of technical skills, the need for continuous improvement and gaining input from a wide array of stakeholders, such as subcontractors, clients, and end-users, makes development work especially demanding. For example, highlighting the role of effort and commitment to the task at hand in addition to talent, grit has been found to predict success beyond e.g. intellect (Duckworth, *et al*. 2007). However, current education may not be targeting these soft skills sufficiently – although skillful and knowledgeable in their subject matters, the academia has awakened to the fact that for example many graduating engineers lack qualifications that enable performing working life duties (Lappalainen 2009), having shortfalls in communication, decision-making, problem-solving, leadership, emotional intelligence, and social ethics skills (Nair *et al*. 2009). One of the crucial additions to substance expertise is understanding development motivation, and being able to enhance it both in one's own work and in the development project in hand. With the term development motivation we refer to people's general strive to learn, develop themselves, master new skills and apply their talents responsibly according to a standard of excellence (see Ryan & Deci 2000). Motivation to develop one's skills and competencies stems from the type of unset self-belief or mindset one adopts in life: people with growth mindset believe that they can develop themselves, failure is nothing but a valuable learning experience, and challenges are opportunities to develop further. (Dweck 2006.) Adopting this type of mindset is crucial for creativity in where coping with the advances, opportunities, technologies and changes is necessary (Runco 2004). Creativity is usually viewed both as a process and an outcome (Ancona & Caldwell 1993) and can be defined as an approach to work that leads to the generation of novel and appropriate ideas, processes or solutions (Amabile 1996).

Unfortunately, development motivation cannot be secured by mere external compensation. In fact, external rewards often diminish intrinsic motivation (Deci *et al*. 1999; Amabile 2003), foster short-term

thinking, and result in poorer performance especially in creative tasks due to narrowing performers foci (e.g. Glucksberg 1962). Indeed, ensuring initiative and full collaboration in development requires a deeper level of motivation than one based on external rewards (Amabile 1998; Simonton, 1999). Instead of promoting extrinsic motivation stemming from the desirability of external consequences of the task, the focus should be on intrinsic motivation stemming from the desirability of the task itself (Ryan & Deci 2000). Intrinsic motivation refers to the inherent tendency of humans to seek out challenges, and to extend and exercise one's capacity (Ryan & Deci 2000). Several motivation theories and models agree on the significant effect that perceived meaningfulness, accomplishment and control have on motivation (Amabile *et al.* 1996; Deci 1975; Gagné & Deci 2005; Hackman & Oldham 1976; Ryan & Deci 2000; Thomas & Velthouse 1990). Intrinsic motivation is especially beneficial for creativity (Amabile 1996), and is frequently included as a core characteristic of creative persons in personality research (Runco 2004). Creativity can occur in diverse situations and result in the generation of new ways to perform work, coming up with novel procedures or innovative ideas, and reconfiguring known approaches into new alternatives (Perry-Smith & Shalley 2003). Thus intrinsic motivation and the ways of promoting and sustaining it clearly become the most crucial components of enhancing development will in creative design work. The question of how to enhance intrinsic development motivation, however, can be a daunting one to answer, as the literature is dispersed, partial answers being presented in multiple fields of research.

In addition to intrinsic motivation, the role of experimentation in design has long been recognized amongst research and practitioners alike. Experimentation is an iterative hands-on process that fosters taking initiative and includes numerous rounds of trial and error (Brown 2008). Further, the process is to a certain extent open-ended and uncertain (Lawson 2005). This experimental approach can be manifested in several ways of which the presence of prototypes, or "approximation[s] of the product along one or more dimensions of interest" (Ulrich & Eppinger 2000, p. 275), is probably the best evidence of experimentation occurring in an organization (Brown 2009). Continuous experimentation through prototypes provides a tool that can be used to explore ideas and stimulate thinking (Koria *et al.* 2011), and openness to experimentation is essential to any creative organization (Brown 2009). For example, models and simulations are used to reflect, improvise, and evaluate generated ideas (Brown 2008).

Much research has been conducted on experimentation from variety viewpoints, such as the effects of experimentation on knowledge generation (West & Iansiti 2003), the effect of model building on idea generation and evaluation (Lemons *et al.* 2010) and novel experimental methods (e.g. Thomke, von Hippel & Franke 1998). Experimentation is also a central feature in the now-booming design thinking literature (Brown 2008; Clark & Smith 2008; Hassi & Laakso 2011; Schön 1991), both in the design studies discourse as well as in non-design-led disciplines (e.g. Burney & Hyer 2006; Kimbell 2009). However, studies have tended to focus on the effects of experimentation on the quality of the outcome, rather than the psychological effects of the practice. The aim of this paper is to shed light on how experimentation can be utilized to enhance development motivation. Literature on experimentation and motivation is reviewed from several streams of research, including psychology, design thinking, organizational creativity, management, and product development, explicating the basic mechanisms through which experimentation can be utilized in order to have a positive effect on intrinsic development motivation. After discussing the basis of these leverages, we present initial findings from a recently developed teaching platform, Aalto University Design Factory, and the role that experimentation has played in the setting. Ultimately this line of research can afford the development of higher quality teaching of design, ensuring the application of subject expertise in the workplace.

THE MOTIVATIONAL EFFECTS OF EXPERIMENTATION

Although a staple practice in design, little attention has been paid to the motivational effects of experimentation. Thus experimentation might be underutilized, overlooking potential psychological

benefits of the practice. After reviewing literature on experimentation and motivation, findings are presented under two main immediate mechanisms: lowering the threshold of participation (by for example providing a common language between different parties, reducing risks, and providing enjoyment) and demonstrating the (potential) effect of efforts (by for example concreticizing the problem or need at hand, providing small wins and demonstrating progress). The implications and the motivational basis of each effect are reviewed in more detail.

LOWERING THE THRESHOLD OF PARTICIPATION

The first key effect of experimentation lies in lowering the threshold of participating in design efforts. As the design process is inherently uncertain, marked by vague starting points, options and goals (Rittel & Webber 1973, Buchanan 1992, West Churchman 1967), participating in the progress requires some courage. Experimentation can help to lower the threshold via five mechanisms: lowering what is at stake, reducing risks, taking small steps, making communication easier, and providing an enjoyable experience.

Aiming for innovations, risk taking is a necessity in development work. However, early experimentation with several ideas (Sarasvathy 2003) and short-term experiments (Chandler et al. 2009) can be utilized to reduce risks to affordable levels. The faster ideas are made tangible, the sooner it is possible to evaluate them and choosing the best solution (Brown 2009). Early prototyping as a one form of experimentation helps to avoid costly mistakes (Brown 2009), and mistakes can be learned from. By decreasing the level of risk associated with the development, the attractiveness on engaging in development efforts can be increased by experimentations (Brown 2009; Zaccai in Lockwood 2010). By adopting different forms of experimentation practitioners promote control in the face of uncertainty (Gerber 2009) and reframe failure as acceptable and necessary rather than something to be avoided (Gerber & Carroll 2011).

Another crucial aspect of experimentation in motivating design work is that it lowers the threshold for participation and acceptance, as only small steps are taken. By experimentation, larger tasks can be divided into modest size tasks allowing frequent action (Gerber & Carroll 2011). In a classic study on the foot-in-the-door technique, Freedman and Fraser (1966) found that while the majority of people refused to place a large sign on their lawn if requested directly, but first asking for the placement of a small sign and then later on a large one had a high level of compliance. One of the reasons behind this phenomena is making decisions based on identity rather than consequences (March 1994): accepting the initial small request based on neutral consequences may mold the person's perception of themselves so that the next time when they are approached with a request, the answer to the question "what would a person like me do" has become favorable to the cause, and the cost of collaboration is no longer the key issue. Therefore small experimentations are helpful in introducing new, even radical ideas into a design process, as they lower the threshold of acceptance. Thus experimentation affects development will not only via intrinsic motivation, but a number of other mechanisms as well.

Experimentation can also ease collaboration between different stakeholders. One of the central features of experimentation is the extensive use of visualizations and prototypes. Indeed, designers almost always draw and frequently construct prototypes (Lawson 2005), as a method of stimulating thinking and exploring ideas (Boland & Collopy 2004). Prototypes can be created on a continuum of formats ranging from concept sketches to preproduction versions of a product (Ulrich & Eppinger 2000), but rather than producing final solutions, the goal is to learn from the experiments and identify new directions for the process (Brown 2008). These prototypes also serve as communication tools. It is especially important for development motivation for one to understand the concepts and ideas pursued, in order for efforts to be perceived as meaningful. The broader the community of stakeholders grows to, the more crucial the role of physical objects that bridge boundaries becomes (Leonard-Barton 1995). Using prototypes helps to create a shared vision in a situation where multiple people are working on the same problem, and prevents miscommunication (von Stamm 2003) both inside the development team and especially when

dealing with other stakeholders such as users. It takes less effort to discuss and share ideas (Junginger 2007) when visualizations are used as a tool for communicating and gaining common understanding on the topic (Ward *et al*. 2009). Thus the burden of taking part in development activities is minimized for collaborators, again lowering the threshold of expending the required amount of effort.

Finally, the role of enjoyment should not be belittled. Experimentation, such as prototyping and building models is a practice frequently reported as enjoyable by both students and practitioners. Positive emotions, in turn, do not only feel good at the moment, but build future capacity (Fredrickson 2001). Unlike negative emotions that ignite immediate behavioral changes to deal with the current situation (Fredrickson 1998), current positive emotions can promote future resilience in the case of setbacks (Fredrickson *et al*. 2003; Waugh *et al*. 2008). As these are bound to occur in development work, increasing willingness to persist in expending efforts even when faced with setbacks increases the likelihood of successful development results. Also particularly significant for creative work is the broadening effect that positive emotions have: attentional scope increases to enhance perceiving the "big picture" (Fredrickson & Branigan 2005). Employees become better at integrating information (Isen *et al*. 1991), thought-action repertoires become more numerous (Fredrickson & Branigan 2005) and fixating on initial ideas becomes less likely (Isen *et al*. 1991). In addition to being directly beneficial for the design result, a broadened perspective can help to increase perceptions of relatedness, thus increasing motivation (Ryan & Deci 2000).

DEMONSTRATING THE EFFECTS OF EFFORTS

The second key effect that experimentation produces is that of demonstrating the impact of efforts. The significance of demonstrating progress has lately been highlighted in development work. For example, based on 12 000 diary entries, Amabile and Kramer (2011) report progress as the single largest contributor towards day-to-day motivation in creative work. Demonstrating progress can increase perceptions of capability to influence the situation, thus affecting both the control and competence parameters of intrinsic motivation. Three aspects of experimentation are considered here: concretizing the idea and problem, producing small wins and providing feedback that enhances learning.

Firstly, prototyping makes concepts concrete and ideas tangible so that they can be shared and evaluated in the real-world settings (Koria *et al*. 2011). Being judged based on the solutions' value-in-use, prototypes can suggest alterations to frameworks, principles and even the intent behind the effort (Sato *et al*. 2010). As constrains are seen as positive challenges (Koria et al. 2011) and problems are made tangible (Sato *et al*. 2010), feelings of personal mastery and competence, as well as relatedness, are enhanced, which in turn have a positive effect on motivation (Ryan & Deci 2000).

A second crucial aspect of experimentation is that it enables producing small wins, scaling down what is at stake (Reay *et al*. 2006), and offering immediacy, tangibility, and controllability (Weick 1986). Small wins can motivate design work by attracting allies, deterring opponents, and lowering resistance to subsequent proposals (Weick 1986), as well as introducing early confidence (Weick 2001), increasing self-efficacy (Weick 1984) and promoting learning (Hollander 1965). For example, Reay, Golden-Biddle and Germann (2006) studied the introduction of a new role to Canadian health care, finding that small wins played a significant role in the process. Small wins marked progress, generated confidence, energized continued work towards the cause, and shifted attention towards the next necessary steps. They helped others begin to see more concretely what the new role would look like in everyday work and gain confidence that the new role was a good alternative to the existing ways of delivering care. In effect, the series of small wins had an escalating nature, and they collectively legitimized change. Small wins can thus promote competence and perceptions of autonomy or control over the situation, both key components of self-motivated behavior (Gagné & Deci 2005; Ryan & Deci 2000). In fact, creating small wins can be adopted as a strategy, in which people "identify a series of controllable opportunities of modest size that produce visible results" (Weick 2001, p. 427). Cultivating such a strategy can promote commitment

and motivation by infusing situations with meaning, reinforce the perception of control or ability to influence the situation, create challenges that is manageable in size, and if continued, can even build up resistance to stress (Weick 1986).

Finally, in addition to producing small wins that mark progress, experimentation can enhance learning through the rapid feedback it provides (Thomke 2003). Experimentation is an iterative process where one learns about the feasibility of the idea through success and failure, with failures often hinting at what will work and which path should be followed (Sato *et al*. 2010). In fact, experimentation can be called practice-based learning (Sato *et al*. 2010), helping to build expertise. Perceiving one's self as capable, in turn, is crucial for intrinsic motivation (e.g. Ryan & Deci 2000)

In conclusion, in addition to enhancing design quality, experimentation is a crucial tool for creating and sustaining motivation through its ability of lowering the threshold for participation and demonstrating progress. Through these two mechanisms, experimentation impacts perceived capability, progress and control over the situation, all three central tenets of intrinsic motivational models (Amabile *et al*. 1996; Deci 1975; Gagné & Deci 2005; Hackman & Oldham 1976; Ryan & Deci 2000; Thomas & Velthouse 1990). Next we introduce a recent university platform development built on promoting experimentation, and discuss the initial findings of the impact it has had on the behavior of students, teachers, researchers and entrepreneurs.

AALTO UNIVERSITY DESIGN FACTORY: AN EXAMPLE OF CREATING A PLATFORM ENHANCING CO-CREATION

To illustrate how an environment built to encourage and support experimentation in development work has been perceived by its users, we discuss the case example of the recently developed Aalto University Design Factory (ADF) platform, a 4000 square-meter facility on one of the Aalto University campuses. In the beginning of 2010, a new university was formed in Finland merging together the nation's three leading universities of technology, art and design, and economics. The new institution, named Aalto University, has aimed to open up new possibilities for strong multidisciplinary education and research, creating a "unique, integrated seedbed for innovation" (Green 2009). The merger has been seen as a flagship project in the larger scale development of the higher education and innovation systems in Finland and essentially as a symbol of the holistic national-level approach to innovation (Kao 2009). One of the new entities created in 2008 in preparation of the new university was Aalto Design Factory, in essence a platform for interdisciplinary co-creation intended to confront some of the widely recognized shortfalls in engineering education, such as taking into account so-called soft skills (e.g. Nair *et al*. 2009; Lappalainen 2009) and motivational aspects that are often poorly present in the traditionally single-disciplinary and single-subject engineering courses. There is ample evidence that students learn most effectively when they understand and believe in their need to master the studied issues. Therefore establishing context, relevance and a linkage to a larger whole is crucial for motivation. This view is widely supported in literature on cognitive and educational psychology and effective pedagogy (Feder *et al*. 2000). In addition to students and teachers, the platform hosts researchers, entrepreneurs and industry representatives. During the academic years of 2008-2009 and 2009-2010, ADF hosted 32 courses for approximately 900 bachelor- and master-degree-level students, 12 research teams, 7 collaboration corporations, as well as several training activities for both academics and practitioners, and numerous start-up companies.

CONTINUALLY DEVELOPING EXPERIMENTATIONAL PLATFORM

Experimentation lies at the heart of the ADF platform, and its development has been guided by four interconnected principles: *crossing boundaries in co-operation*, *lowering the threshold for experimentation - promoting hands-on doing and prototyping*, *increasing initiative and enthusiasm*, and *cultivating an open climate*. Although the support forms develop continuously in an iterative and expansive manner, the underlying basic philosophy has remained the same. There are three key considerations in promoting experimentation amongst the platform users, namely encouraging an non-hierarchical atmosphere and taking initiative,

promoting an open atmosphere and brining together different groups of people, and providing easy access to prototyping tools and materials.

Aalto Design Factory provides an extremely non-hierarchical, constantly developing collaboration environment for students, researchers and business practitioners across hierarchical, professional, and disciplinary boundaries. For example, there are no separate facilities or offices for management but everyone uses the same premises. As earlier studies have showed, organic structures, characterized by lack of hierarchies, low levels of bureaucracy, flexibility, and adaptability, support the climate of trial-and-error (Van deer Panne *et al*. 2003). Furthermore, organic structures increase successful collaboration as they are more conductive for interdepartmental communication and learning, supporting the development work (Monaert *et al*. 2000). This is further enhanced by bringing together designers and non-designers, as well as students, researchers and business representatives, to the same physical facilities enabling diversity in perspectives and creative problem solving. The spatial solutions encourage in communication and collaboration, both of the premeditated and serendipitous type. The facilities also promote user initiative, as all spaces are made flexible and have to be set up by the user at the beginning of a session. Finally, prototyping is encouraged by providing diverse facilities for prototyping, such as hobby materials for rough prototypes, mechanical and electrical proto-shops for more advanced prototypes, teaching the advantages of prototyping in the courses held in the Factory, and inspiring its users by displaying prototyped made by its users, for example in the form of different galas. With its informal and non-hierarchical environment and flexible spatial solutions encouraging interaction, taking initiative and prototyping, the Aalto Design Factory aims to support world-class development in the contexts of education, research and practical application.

PROVOKING DEVELOPMENT MOTIVATION AND CO-CREATION AT THE AALTO DESIGN FACTORY

To gain insight on the impact of the experimentation platform and how Aalto Design Factory is perceived by its users, data was gathered in two anonymous surveys conducted in 2009 and 2010. Altogether 55 respondents consisting of students, researchers and staff members working at the Design Factory on a daily basis or visiting several times a week answered the questionnaires. The data was analyzed adapting a content analysis methodology. The focus was on the core elements behind the concept of the Aalto Design Factory and its role in provoking and sustaining development motivation amongst its users. The respondents of the study were asked if they were encouraged somehow by the ADF facility. The resulting categories are presented in table 1. The users perceived the main effect of the platform to be expanding boundaries. The answers forming the category reflected how the users had perceived the Aalto Design Factory as encouraging them to aim high and try harder. As one respondent put it, ADF encourages its users to “try their wings” and get involved with something they haven’t done before. The study provided justification for labeling the platform as enhancing experimentation.

Every sixth respondent felt that, in comparison to other environments, the ADF platform encourages experimentation. The respondents reported that ADF had encouraged them to have a more hands-on approach and to try things out and experiment instead of only thinking. As one respondent put it, the platform encourages to “*turn ideas into practical work*”. In addition, several respondents mentioned the atmosphere of the ADF platform to give them more courage and to help them trust themselves more. *“Design Factory encourages me to be myself and trust in myself.”*

Further, the respondents emphasized the positive effect of the ADF platform on collaboration. ADF was perceived to support creating efficient networks among professionals from different disciplines and sharing different point of views as well as question one’s own opinions. As one responded noted: *“Design Factory encourages me to challenge my opinions with other people.”*

The last category, learning, comprised of notions the platform encouraging its users to learn new things and to understand how and why things work in a certain way. It was mentioned that Design Factory has taught its users to combine things and to perceive the big picture more easily.

Expanding one's boundaries	40 %
Experimentation	17 %
Collaboration	17 %
Learning	13 %

Table 1. Implications of how Design Factory encourages its users

When asked if the respondents had changed or developed some way with the help of the ADF platform, the two most common answers were that the platform had helped to develop their thinking (16%) and increased their confidence (12%). The respondents perceived that ADF platform had opened their minds to new ways of working, and taught them to be more open towards different goals and different point of views. *"I have learned to think more openly about new domains and see things from different points of views."* On the other hand, the respondents also saw that they had come more tolerant towards uncertainty, were in a sense braver than before, and that they had learned to communicate in a more straightforward manner. *"I can tolerate uncertainty better and I am braver than when I first came here (Design Factory)."* These results are presented in table 2.

Development of thinking	16 %
Increase in the self-confidence	12 %
Broadening one's perspective	<10 %
Increase in the tolerance of uncertainty	
Increase in the courage	
Increase in the straightforward and open communication	

Table 2. Implications of how Design Factory affects its users

In general, openness, communal and inspired atmosphere and the possibility to work with like-minded and interesting people were highly valued among the respondents in the case environment that puts emphasis on experimentation and continuous development. On the other hand, dispersion of the community, hurry, lack of information and technical difficulties due to the missing technical support were seen as disadvantages of the platform. ADF was perceived to differ from other work or study environments in terms of its informality, its inspiring and permissive atmosphere, interaction promoting spaces, and by the possibility to meet interesting people. Finally, the respondents described Design Factory as an inspiring, fun, open and free place.

The results thus presented provide preliminary observations on how a platform that underlines the importance of experimentation is perceived by its users and what kinds of effects it can have on its users. In addition to encouraging experimentation, the platform was perceived to broaden boundaries (increasing motivation as the big picture became more clear) and promote collaboration, increasing willingness to conduct development work with a wide variety of people. In addition, the platform was perceived by its users to enhance learning, develop their thinking and provide them with more confidence, increasing perceptions of self-efficacy (i.e. the expectancy that one is able to perform successfully, Bandura 1982) and thus further enhancing intrinsic motivation to engage in development. Last, but not least, the platform was simply perceived as inspiring and fun. Although future research is required on the platform, the initial findings suggest that supporting experimentation can indeed enhance development will and provide a fruitful focus for educational efforts.

CONCLUSIONS AND IMPLICATIONS

Motivation plays a crucial role in development, influencing the extent of skill utilization, collaboration and persistency, among other factors. Yet it is far from clear how motivation should be supported in design work, and most training still focuses on technical mastery. On the other hand, experimentation has long been recognized as a central practice in design and development work. As the practice is already widespread, experimentation may offer a fruitful starting point for enhancing motivation. Although examined from numerous angles, previous research has tended to focus more on the effects that experimentation has on the quality of the design output, rather than impact of experimentation on design motivation. This paper reviewed a wide array of literature, explicating the effects that experimentation can have on motivation

through lowering the threshold of participation and demonstrating progress through various mechanisms. The findings, complemented with their implications at the case presented, are reviewed in Table 3.

Although the importance of experimentation in design context has been recognized in earlier studies, this study presented a unique view on a case, Aalto University Design Factory, explicitly aiming to benefit design work through promoting experimentational approach. Based on initial findings from two questionnaires completed by the ADF platform users, the positive effect that the platform is perceived to have on experimentation was confirmed. The results emphasized how important the users experienced the communal atmosphere that encourages one to learn and gives a freedom to explore. Indeed, Baer and Frese (2003) have highlighted the significant role of the organizational climate in creating innovations. The ADF platform users (students, teachers, researchers and entrepreneurs) felt that they had possibilities to challenge themselves, and perceived the array of multidisciplinary people that the platform had brought together as something that helped and supported them in the challenges they engaged in. Finally, the platform was perceived by its users to enhance learning, develop thinking and increase confidence.

The fact that all the interviewees were internal members of ADF has an effect on the objectivity of the study. However, since the study is a case study, analysis of an individual unit (ADF) is reasonable. Although the empirical basis of the paper is merely illustrative and more research is yet needed, the review of existing research as well as the initial findings of the ADF platform suggest that supporting experimentation can indeed enhance development will and provide a fruitful focus for future educational development efforts.

Main mechanism: Demonstrating progress		
Key feature	**Effect on motivation**	**Implications at the ADF**
Starting with small experimen-tations, proceeding to larger ones	Lowering the threshold of acceptance (Freedman & Fraser 1966) and transitioning from cost to identity based decision making (March 1994)	Design Factory was perceived to encourage experimentation
Providing a common language with prototypes	Creating a shared vision ensuring the meaningfulness of contributions, lowering the amount of effort required for communication (Leonard-Barton 1995; von Stamm 2003; Junginger 2007; Ward et a. 2009)	Design Factory was perceived to support creating efficient networks among professionals from different disciplines and sharing different point of views
Positive emotions stemming from prototyping and concrete doing	Increased perceptions of relatedness (a key component of intrinsic motivation) due to a broadened perspective, increased resilience (Fredrickson & Branigan 2005; Ryan & Deci 2000)	Design Factory was perceived to broaden boundaries as well as perspective
Main mechanism: Lowering the threshold for participation		
Key feature	**Effect on motivation**	**Implications at the ADF**
Concretizing the design problem and solution by prototypes	As constrains are seen as positive challenges and problems are made tangible, feelings of personal mastery and competence, as well as relatedness, are highlighted (Ryan & Deci 2000; Koria et al 2011; Sato et al. 2010).	Design Factory was perceived to encourage hands-on approach as well as thinking-by-doing
Producing small wins	Promoting competence and perceptions of autonomy or control over the situation, two of the three key components of self-motivated behavior (Gagné & Deci 2005, Ryan & Deci 2000).	Design Factory was perceived to increase perceptions of self-efficacy
Providing rapid feedback	Enhancing learning and thus increasing perceptions of capability (Thomke 2003), resulting in greater intrinsic motivation.	Design Factory was recognized to increase more straightforward communication and perceptions of self-efficacy

Table 3. Key effects of experimentation on development motivation

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